S-PERFECTNESS OF A PSEUDO-COHERENT COMPLEX IS DETECTED ON THE FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. Inputs lemma-affine-locally-rel-perfect and lemma-compare-base-change are same-chapter NOT carded (cross-line-claimed labels) and are declared external.

Allowed standard prerequisites (editorial).

- Affine-local nature of relative perfectness. lemma-affine-locally-rel-perfect (same chapter, NOT carded, declared external). Reduces to X and S affine.
- Fibre pullback as derived tensor with the residue field. lemma-compare-base-change (same chapter, NOT carded, declared external). For flat A -> R, the pullback to X_s corresponds to K tensor^L_R kappa(p).
- Algebra criterion for boundedness on fibres. More on Algebra, lemma bounded-on-fibres-relatively-perfect (cross-chapter, named). The corresponding module-level statement.

Source reference. Original lemma 0DI2: sources/stacks-perfect/pinned-original-source.tex, source lines 8951–8951. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 08IB: sources/stacks-perfect/pinned-original-source.tex, source lines 5362–5362. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0GHJ: focused-seeds/geometry/S9-perf2-bounded-on-fibres/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes which is flat and locally of finite presentation. Let E be a pseudo-coherent object of $D(\mathcal{O}_X)$. The following are equivalent*

(1) *E is S-perfect, and*
(2) *E is locally bounded below and for every point $s \in S$ the object $L(X_s \to X)^*E$ of $D(\mathcal{O}_{X_s})$ is locally bounded below.*

Date: stacks-project:perfect.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Since everything is local we immediately reduce to the case that X and S are affine, see Lemma 0DI2. Say $X \to S$ corresponds to $\mathrm{Spec}(A) \to \mathrm{Spec}(R)$ and E corresponds to K in $D(A)$. If s corresponds to the prime $\mathfrak{p} \subset R$, then $L(X_s \to X)^*E$ corresponds to $K \otimes^{\mathbf{L}}_R \kappa(\mathfrak{p})$ as $R \to A$ is flat, see for example Lemma 08IB. Thus we see that our lemma follows from the corresponding algebra result, see More on Algebra, Lemma 0GHJ. $\square$